

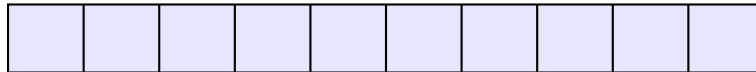
Finding a Percent of a Quantity

Benchmark percents on a hundred grid and a bar model, the decimal multiplier, the fraction shortcut, and percent applications

How to work:

- “Percent” means **out of one hundred**. A percent of a quantity is a part of that quantity, so 30% of 70 means $\frac{30}{100}$ of 70.
- Three routes give the same answer: split the whole on a **bar model**, multiply by the **decimal**, or multiply by the **fraction**.
- Every problem here asks for money in dollars, a count, or a number — no units are hidden in the arithmetic.
- Show the route you used, then write your answer on the line.

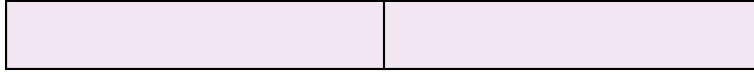
1. Bar model. The whole bar below stands for 80 books. It is divided into ten equal parts, so each part is 10% of the books.



Shade one part. How many books is 10% of 80? Write the number inside the part you shaded.

Answer: _____

- 2. Bar model.** The bar below stands for 46 marbles.



The bar is split into two equal parts. Explain what one part shows about 50%, then find 50% of 46.

Answer: _____

- 3. Ratio table.** Fill in the table to find 30% of 70.

Percent	Amount
100%	70
10%	
30%	

Then write the multiplication that gives the same answer in one step.

Answer: _____

4. Fraction shortcut. A benchmark percent can be replaced by a simple fraction: 25% of an amount is the same as $\frac{1}{4}$ of it. Use the fraction to find 25% of a playlist of 64 songs, and show the division you used. Then check your answer by multiplying by the decimal instead.

Answer: _____

5. Find the mistake. Dana wants 5% of 120. She moves the decimal point one place to the left and writes 12.

- Which percent did Dana actually find? Explain how you know.
- Write the correct value of 5% of 120.

Answer: _____

6. Restaurant tip. A meal costs 48 dollars and the family leaves a tip of 15% of the bill. How many dollars is the tip? Give your answer to the nearest cent.

Answer: _____

7. Concert hall. A hall has 250 seats. On Tuesday 8% of the seats were empty. How many seats were empty?

Answer: _____ seats

8. Jacket on sale. A jacket costs 35 dollars and is marked 20% off.

- Let d be the discount in dollars. Write an equation for d and solve it.
- Use your discount to find the sale price in dollars.

Answer: _____

9. An awkward percent. A recipe calls for 12.5% of a 64-gram block of chocolate. 12.5% is the benchmark fraction $\frac{1}{8}$. Find the mass called for, and explain why the fraction route is easier here than the decimal route.

Answer: _____

10. Two percents in a row. A backpack costs 60 dollars. It is marked 25% off, and then 8% sales tax is added *to the sale price*.

- Let s be the sale price in dollars. Write an equation for s and solve it.
- Find the final price in dollars, to the nearest cent.
- Say why the tax is not taken on the original price.

Answer: _____